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Quick Start

The example code provided in this quick start guide is for educational and demonstration purposes only. It may not represent best practices for production use.

Breaking Changes Notice

If you've just updated the package, it is recommended to check the [changelogs](#) for information on breaking changes.

Prerequisites

- Unity Version: Unity 6.0+ (tested on Unity 6.3).
- Importer Dependencies (Install at least one):
 - [glTFast](#) for GLTF/GLB support.
 - [UniGLTF](#) for alternative GLTF/GLB support.
 - [UniVRM](#) for VRM support.
- Other Optional Dependencies:
 - Unity's Animation module for animator post-processing.

Key Concepts

- **Data Loaders:** Fetch raw avatar data (model, metadata, renders) from files or URIs.
- **Importers:** Convert raw data into Unity GameObjects.
- **Post-Processors:** Customize loaded avatars (e.g., add animators, override materials).
- **AvFlow:** Chains loaders, importers, and post-processors for a complete loading pipeline.

Avatars require a specific structure:

- Model file (e.g., `model.glb`, `model.vrm`).
- `metadata.json` with ID, type, gender, etc.
- Optional renders: `full.jpg/png`, `bust.jpg/png`.

Basic Usage

Use `AvFlow` to build a loading pipeline.

Example: Load from Local File

```
public class AvatarLoader : MonoBehaviour
{
    async void Start()
    {
        // FileAvDataLoader supports auto-detection of the model format by file extension
    }
}
```

```

var flow = new AvFlow.Builder()
    .From(new FileAvDataLoader("path/to/avatar/dir")) // Directory with model,
metadata, renders
    .Using(new GLTFastAvImporter()) // Or UniGLTFAvImporter, UniVRM10AvImporter
    .Build();

LoadedAv? avatar = await flow.TryRunAsync(); // Or `RunAsync()` for explicit
throwing errors.
if (avatar != null)
{
    avatar.GameObject.transform.position = Vector3.zero; // Position in scene
    // Use avatar.GameObject, avatar.Metadata, etc.
}
else
{
    Debug.LogError("Failed to load avatar");
}
}
}

```

Example: Load from URI (Remote)

```

var flow = new AvFlow.Builder()
    .From(new URIAvDataLoader(
        "https://example.com/model.glb",
        "https://example.com/metadata.json",
        AvModelFileExtension.GLB // Model format must be explicitly declared
    ))
    .Using(new GLTFastAvImporter())
    .Build();

LoadedAv avatar = await flow.RunAsync(); // Throws on failure

```

Adding Post-Processors

Post-processors run after import. Synchronous ones run first, then async.

```

// In AvFlow.Builder
.ProcessWith(new AnimatorPostProcessor() { AnimatorController = myController }) //
Add animator
.ProcessWithAsync(new CachePostProcessor("cache/dir")) // Cache downloaded model to
local dir

```

- **Material Overrides/Shader:** Use `MaterialOverridePostProcessor` or `ShaderSwapPostProcessor` with configs (create via ScriptableObject).
- **Conditional:** Wrap with `ConditionalPostProcessor` for logic-based processing.

Sync processors (e.g., `AnimatorPostProcessor`) always run before async ones (e.g., `CachePostProcessor`), regardless of addition order.

The `AvPost` class has static methods for fluent-style creation of post-processors.

Multi-Importer Example with More Post-Processors

```
[SerializeField] RuntimeAnimatorController _animator;
[SerializeField] Avatar _femAvatar;
[SerializeField] Avatar _mascAvatar;
[SerializeField] bool shouldLoadGltf;

new AvFlow.Builder()
.From(
    new FileAvDataLoader(
        shouldLoadGltf
        ? Path.Join(Application.streamingAssetsPath, "GLTFModelFolder")
        : Path.Join(Application.streamingAssetsPath, "VRMModelFolder")
    )
).Using(
    new UniVRM10AvImporter(),
    new GLTFFastAvImporter(),
    new UniGLTFAvImporter()
    {
        ImporterContextSettings = new UniGLTF.ImporterContextSettings(invertAxis:
UniGLTF.Axes.X) // For compatability with glTFast
    }
).ProcessWith(
    AvPost.ConfigureAnimator(_animator, _mascAvatar, new Dictionary<AvGender, Avatar>()
    {
        { AvGender.Feminine, _femAvatar },
        { AvGender.Masculine, _mascAvatar },
    })),

    AvPost.WhenLoadedWith<FileAvDataLoader>(
        AvPost.Do(static (_, _) => print("LOADED AVATAR FROM FILES"))
    ),
    AvPost.WhenLoadedWith<URIAvDataLoader>(
        AvPost.Do(static (_, _) => print("LOADED AVATAR FROM URIS"))
    ),

    AvPost.WhenImportedWith<UniVRM10AvImporter>(
```

```

        AvPost.Do(static (_, _) => print("LOADED VRM AVATAR"))
    ),
    AvPost.WhenImportedWith<UniGLTFImporter>(
        AvPost.Do(static (_, _) => print("LOADED UNIGLTF AVATAR"))
    ),
    AvPost.WhenImportedWith<GLTFastImporter>(
        AvPost.Do(static (_, _) => print("LOADED GLTFAST AVATAR"))
    )
).ProcessWithAsync(new CachePostProcessor(Application.temporaryCachePath))
.Build();

```

Handling Renders and Metadata

- Access `avatar.FullRender` and `avatar.BustRender` for thumbnails.
- `avatar.Metadata` includes ID, gender, type (e.g., `AvType.HumanoidFullBody`).

Cleanup

Always dispose loaded avatars:

```

avatar.Dispose(); // Destroys GameObject, renders, post-processor created materials, etc.

```

For full API docs, please see <https://uralstech.github.io/AvLoader/api/Uralstech.AvLoader.html> or `APIReferenceManual.pdf` in the package documentation for the reference manual.

You can implement `IAvDataLoader` or `IAvImporter` for custom sources/importers (e.g., cloud APIs, FBX import).